

Nonnegative and positive factorizations

The nonnegative rank of the nine-point distance matrix

Difficulty: hard **Importance:** interesting to specialist

Status: Solved

Rating rationale: Hard because a single small exact factorization must be constructed or excluded using sharp nonnegative-rank tools; specialist importance reflects its role as a named distance-matrix benchmark.

Area: exact factorization of Euclidean distance matrices

Resolution — 2026-09-11

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The nine-point matrix $D_{ij} = (i - j)^2$ has nonnegative rank seven, so no exact six-term nonnegative factorization exists. A polygon-contact argument applied to both factors, together with Sylvester's rank inequality, proves the lower bound; an explicit seven-term integer factorization proves the upper bound.

The complete target is resolved. Its former difficulty rating is historical; the original statement, references and dated audits remain below.

Primary reference: [complete authored PDF](#), [standalone TeX](#), **Theorem 1**, with **Theorem 4 for the lower bound**. [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Context and notation

All factorizations are over the real numbers. For $X \in \mathbb{R}_{\geq 0}^{m \times n}$, define

$$\text{rank}_+(X) = \min\{r \geq 0 : X = WH, \quad W \in \mathbb{R}_{\geq 0}^{m \times r}, H \in \mathbb{R}_{\geq 0}^{r \times n}\}.$$

Problem statement

Let $D \in \mathbb{R}_{\geq 0}^{9 \times 9}$ have entries $D_{ij} = (i - j)^2$ for $1 \leq i, j \leq 9$.

Question

Does there exist an exact factorization $D = WH$ with $W \in \mathbb{R}_{\geq 0}^{9 \times 6}$ and $H \in \mathbb{R}_{\geq 0}^{6 \times 9}$? Equivalently, is $\text{rank}_+(D) = 6$ or 7 ?

References

Baeckelant, Vandaele, and Gillis, [Computing Lower Bounds on the Nonnegative Rank via Non-Convex Optimization Solvers](#), Appendix A.1, Table 6 and its final paragraph, expressly isolates this case. Pavel Hrubeš, *On the nonnegative rank of distance matrices*, Information Processing Letters **112** (2012), 457–461, supplies the upper-bound construction cited there.

Status check — 2026-09-10

Rechecked [Baeckelant et al. v2](#), [Appendix A.1](#), [Table 6](#) and [final paragraph](#), and searched for later nine-point distance-matrix factorizations. The July 2026 text explicitly retains the gap between six and seven. Its new exact results for orders eleven and twelve do not settle order nine. No six-factor construction or seven-factor lower bound was located.